4 Fundamental Subspaces

Of a Matrix

1) Col space of A | $C(A)$
2) Nullspace of (A) | $N(A)$
3) Rowspace of A | $C(A^T)$
4) Left Nullspace of A | $N(A^T)$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

contribute to colspace.

pivot | free.

$v = 2$ (plane in $\mathbb{R}^3$)

More generalized = Rank.

$$\begin{bmatrix} \boxed{1} & 2 & 3 & 1 \\ 0 & \boxed{-1} & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$



Ran K

	Dimension =	Basis vector
C(A)	r	pivot columns of A.
N(A)	n − r (free cols) total-	Special solutions
C(Aᵀ)	r	pivot Rows of R(A) / A
N(Aᵀ)	m − r	Free rows of Elimination mat E

$$A=\begin{bmatrix}1 & 2 & 3 & 1\\ 1 & 1 & 2 & 1\\ 1 & 2 & 3 & 1\end{bmatrix}$$

$x_3=1$, $x_4=0$ | $x_3=0$, $x_4=1$

↳ $a\begin{bmatrix}1\\ -1\end{bmatrix}$ → backsubstitution

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$a\begin{bmatrix} 1 \\ -1 \\ 1 \\ 0 \end{bmatrix} + b\begin{bmatrix} -1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

$x_{special}$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

pivot Free

$r = 2$

why A^T rank = A rank.

if a col has pivot the row

If a col has pivot then the row also does. & vice versa.

→ Again Transpose :

$$A = \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

row reduced echelon form.

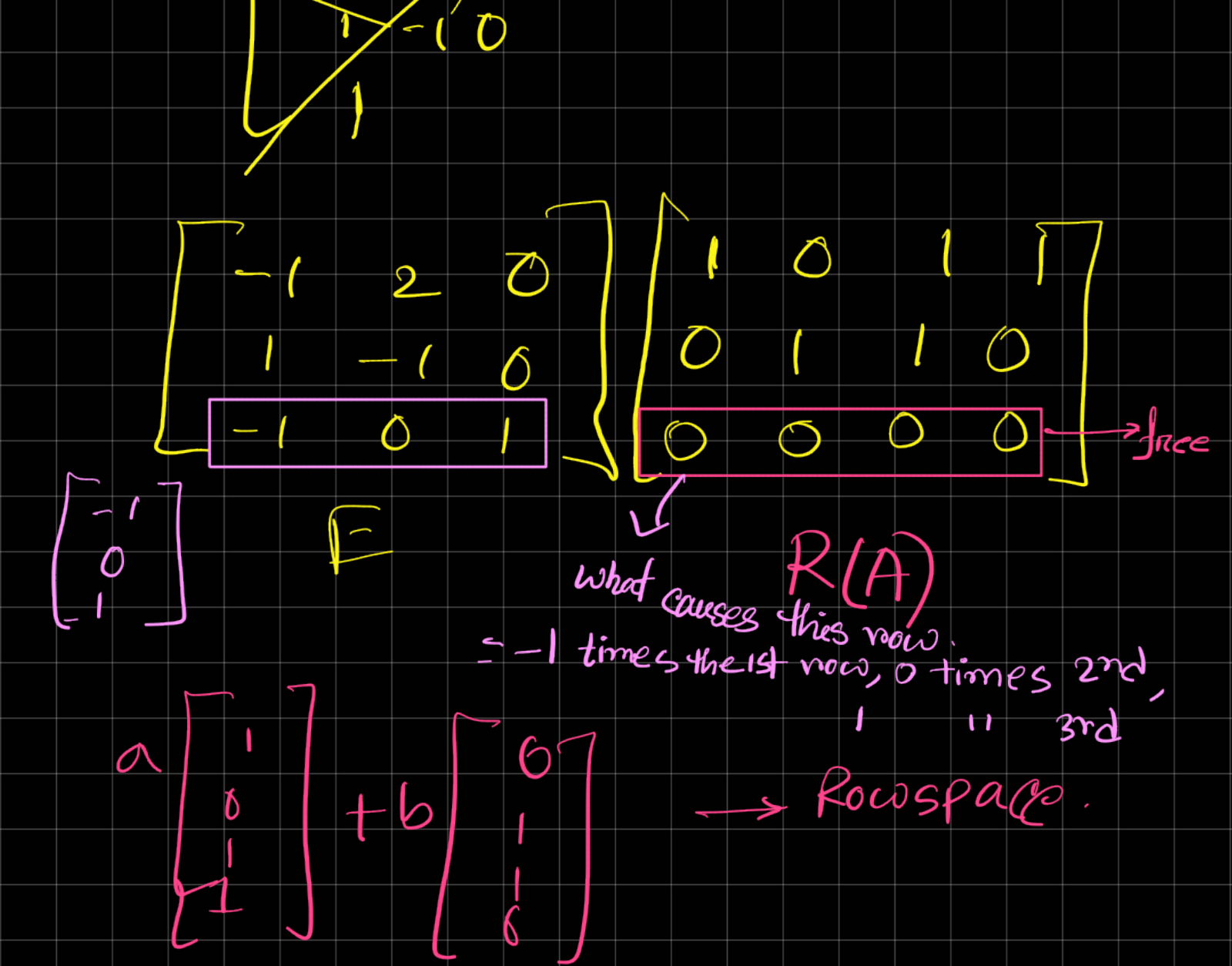
Pivot rows.

Free rows.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 & 1 \\ 1 & 1 & 2 & 1 \\ 1 & 2 & 3 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 3 & 1 \\ 0 & -1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ [1 0 0



$$a\begin{bmatrix}1\\0\\1\end{bmatrix} + b\begin{bmatrix}0\\1\\1\end{bmatrix}$$

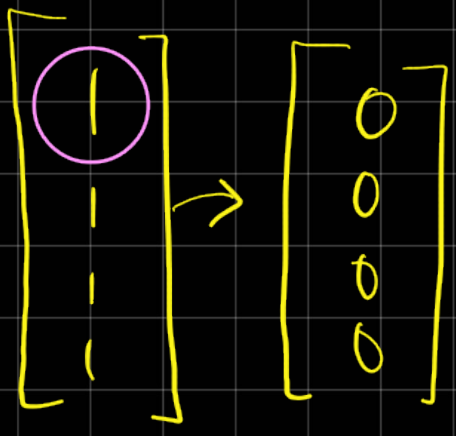
$$C(A) \neq C(R(A)) \quad \text{(For all Mat)}$$

Linear Transformation → any Elimination we work with rows not cols no guarantee colspace would be preserved but row space will be.

$Ux = 0$

$\updownarrow$

$Ux = b$



A

$Ax = 0$ ← Nullspace of A^T = Left Nullspace of A

$A^T y = 0$

$= (A^T y)^T = 0^T$

$$\underset{n \times m}{\begin{bmatrix} & A^T & \end{bmatrix}} \underset{m \times 1}{\begin{bmatrix} y \end{bmatrix}} = \underset{n \times 1}{\begin{bmatrix} 0 \end{bmatrix}}$$

$y^T (A^T)^T$

$= y^T A$

$= [0]_{1 \times n}$

→ Left Nullspace.

$$A^T = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \\ 1 & 1 & 1 \end{bmatrix}$$

pivot, pivot, free

→ Nullspace = A Line

$$E = \left[\begin{array}{c|c} A_{m\times n} & I_{m\times m} \end{array}\right]$$

$$= \left[\begin{array}{c|c} I_{m\times m} & E \end{array}\right]$$

$E \to A^{-1}$

A non square ↓ Non invertible

$$EA = R$$

$$E = I^{-1}$$

$$E = \left[\begin{array}{c|c} A_{m\times n} & I_{m\times m} \end{array}\right]$$

The only luxury for non squares =

$$\left[\begin{array}{c|c} R(A)_{m\times n} & E_{m\times m} \end{array}\right]$$

$E_{m\times m}$ → always square

The closest I can get to I

$$EA = R(A)$$